

Phytate in foods and significance for humans: food sources, intake, processing, bioavailability, protective role and analysis.



The article gives an overview of phytic acid in food and of its significance for human nutrition. It summarises phytate sources in foods and discusses problems of phytic acid/phytate contents of food tables. Data on phytic acid intake are evaluated and daily phytic acid intake depending on food habits is assessed. Degradation of phytate during gastro-intestinal passage is summarised, the mechanism of phytate interacting with minerals and trace elements in the gastro-intestinal chyme described and the pathway of inositol phosphate hydrolysis in the gut presented. The present knowledge of phytate absorption is summarised and discussed. Effects of phytate on mineral and trace element bioavailability are reported and phytate degradation during processing and storage is described. Beneficial activities of dietary phytate such as its effects on calcification and kidney stone formation and on lowering blood glucose and lipids are reported. The antioxidative property of phytic acid and its potential anticancerogenic activities are briefly surveyed. Development of the analysis of phytic acid and other inositol phosphates is described, problems of inositol phosphate determination and detection discussed and the need for standardisation of phytic acid analysis in foods argued.